

PROF. D.S. GAMBHIR

MD, DM, FAMS, FCSI, FICC, FCAPSC, FSCAI, FACC (USA)

PADMA SHRI AWARDEE

GROUP DIRECTOR, CARDIOLOGY, Kailash Health Care Ltd.
DIRECTOR, Kailash Hospital & Heart Institute, Noida 201301

OPENING BLOCKAGE IN THE MAIN CORONARY ARTERY BY STENTING

A Viable Alternative to Bypass Surgery

Nonsurgical treatment of blockage in the arteries supplying blood and oxygen to the heart muscle was first performed in 1977 in Switzerland. Since then millions of patients with this heart disease have been treated all over the world by this non-surgical technique, called angioplasty. It provides relief of symptoms like pain from the heart, and even prolongation of life. Various professional organizations like American College of Cardiology recognized angioplasty as a viable alternative to bypass graft surgery in specific groups of patients. With increasing experience of the operators, availability of better quality of balloons and other ancillary hardwares, the indications of angioplasty have been expanded to treat more complex subset of patients including those with ongoing heart attack, multivessel-multiple blockages and even totally occluded coronary arteries.

Untill mid 70's, bypass graft surgery was the only modality of treatment for patients with drug-resistant angina caused by significant blockage in their main coronary artery which supplies almost 80% of blood to the heart muscle. Sudden complete occlusion of this artery is most often the cause of sudden death, out of hospital cardiac arrest or serious irregularities of heart beat. Although technically easy to approach, balloon angioplasty was not recommended for left main coronary artery blockage because of adverse inhospital and follow-up results. However, with the availability of drug-eluting stents in 2002, which proved their superiority over bare metal stents, treatment of left main coronary artery by angioplasty witnessed a revolution in utilization of this nonsurgical approach.

Favourable results of several non-randomized and observational studies comparing angioplasty with stenting versus bypass surgery paved the way for prospective, randomized trials all over the world. Atleast five multinational trials reaffirmed that implantation of drug-eluting stents for opening the blockage in the left main coronary artery is a viable alternative to open heart bypass surgery in selected group of patients. There is no significant difference in the long-term survival rates between the two treatment modalities. However, patients who undergo stenting require repeat procedure more often as compared to bypass surgery, because of recurrence of blockage within the stent. Even in diabetic patients, both procedures have similar long-term outcome.

The results of stenting in the main coronary artery have been further improved by use of newer technologies like intracoronary imaging with intravascular ultrasound, and optical coherence tomography which enabled viewing the blockage from inside the artery. This provided precise assessment of blockage, selection of stent size and visualization of its proper deployment, thus reducing the future risk of complications and recurrences.

We, at the Kailash Hospital and Heart Institute, Noida are pioneer in this field of nonsurgical treatment of left main coronary blockage. All such procedures at our institution are performed, guided by intravascular imaging, with almost negligible risk of complications and durable long-term results.

The author is the first cardiologist in our country to have performed this procedure almost three decades ago, and have successfully treated more than 650 patients without using scalpel and leaving a scar on patient's chest. However, as a precautionary measure, this sophisticated and highly sensitive procedure should be carried out only by skilled operators after careful selection of patients suitable for this approach. It is equally important to inform the patient and his relatives about both the techniques-surgery versus stenting. An informed consent must be obtained after explaining the merits and limitations of each. No doubt, majority prefer treatment without suffering the pain of surgery and long-term recovery period after surgery. Therefore, removing blockage in the left main coronary artery with stenting is now well recognized and accepted as a viable alternative to bypass surgery in selected subset of patients.

MILESTONES

- Professor, Department of Cardiology, G.B. Pant Hospital, New Delhi for almost 20 years.
- Hon. Editor of Indian Heart Journal (1995-2000)
- Editor 'Cardiology Update' under the aegis' of Cardiological Society of India (2002)
- Published over 125 original papers and authored several books
- Invited to deliver guest lectures, moderate sessions and perform live cases on left main interventions in national and international conferences all over the world
- Developed the first Drug eluting stent in India- second only in the world after USA.
- First to perform angioplasty in left main coronary artery and one of the highest number in India.
- Honorary Fellowship of National Academy of Medical Sciences (FAMS, 2003), India College of Cardiology (FICC, 2005), Cardiological Society of India (FCSI, 2006) and College of Asian Pacific Society of Cardiology (2007)
- Fellowship of the American College of Cardiology (FACC, 2008).
- Fellowship of Society for Cardiovascular Angiography and Interventions, (SCAI, USA 2009)
- President of the Cardiological Society of India (2002)
- Founding members of the Indian College of Cardiology (ICC)
- President of the Indian College of Cardiology (2007-08)
- Life Time Achievement Award from Cardiology Society of India (DB).
- Searle Award from Cardiological Society of India
- "Distinguished Physician Award" from the Association of American Cardiologists of Indian Origin, USA (2002)
- "Life Time Achievement Award" from Delhi Medical Association
- Awarded Pioneer in Interventional Cardiology by Delhi Interventional Council.
- Padma Shri Award by the President of India (2016)